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# AI-Based Phishing Website Detection Using Machine Learning

import pandas as pd
from sklearn.ensemble import RandomForestClassifier


data = {
    'url_length': [20,75,65,18,80,22,60,25,70,19],
    'has_at': [0,1,0,0,1,0,0,0,1,0],
    'num_dots': [1,5,4,2,6,1,4,2,5,1],
    'has_https': [1,0,0,1,0,1,0,1,0,1],
    'num_hyphens': [0,4,3,0,5,0,2,0,3,0],
    'label': [0,1,1,0,1,0,1,0,1,0]
}

df = pd.DataFrame(data)

X = df.drop("label", axis=1)
y = df["label"]


model = RandomForestClassifier(random_state=42)
model.fit(X, y)


def extract_features(url):
    return [[
        len(url),
        1 if "@" in url else 0,
        url.count("."),
        1 if url.startswith("https://") else 0,
        url.count("-")
    ]]


url = input("Enter Website URL: ")

features = extract_features(url)

prediction = model.predict(features)

# Probability
probability = model.predict_proba(features)

phishing_probability = probability[0][1] * 100

# Convert probability to Risk Score
risk_score = round((phishing_probability / 100) * 5, 1)

print("\n========== ANALYSIS REPORT ==========")
print("URL:", url)
print("URL Length:", len(url))
print("Contains @ :", "Yes" if "@" in url else "No")
print("Number of Dots:", url.count("."))
print("HTTPS Enabled:", "Yes" if url.startswith("https://") else "No")
print("Number of Hyphens:", url.count("-"))

print("\nPhishing Probability:", round(phishing_probability,2), "%")
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print("Risk Score:", risk_score, "/5")

if prediction[0] == 1:
    print("\n⚠️ RESULT: PHISHING WEBSITE")
    print("Recommendation: Avoid entering passwords, OTPs or banking details.")
else:
    print("\n✅ RESULT: LEGITIMATE WEBSITE")
    print("Recommendation: Website appears safe.")

print("\n========== ANALYSIS COMPLETED ==========")
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Enter Website URL: http://bank-update@verify-user-secure-login.xyz

========== ANALYSIS REPORT ==========
URL: http://bank-update@verify-user-secure-login.xyz
URL Length: 47
Contains @ : Yes
Number of Dots: 1
HTTPS Enabled: No
Number of Hyphens: 4

Phishing Probability: 80.0 %
Risk Score: 4.0 /5

⚠️ RESULT: PHISHING WEBSITE
Recommendation: Avoid entering passwords, OTPs or banking details.

========== ANALYSIS COMPLETED ==========
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X
  warnings.warn(
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